

# Dimensional modeling

DATA MODELING IN POWER BI

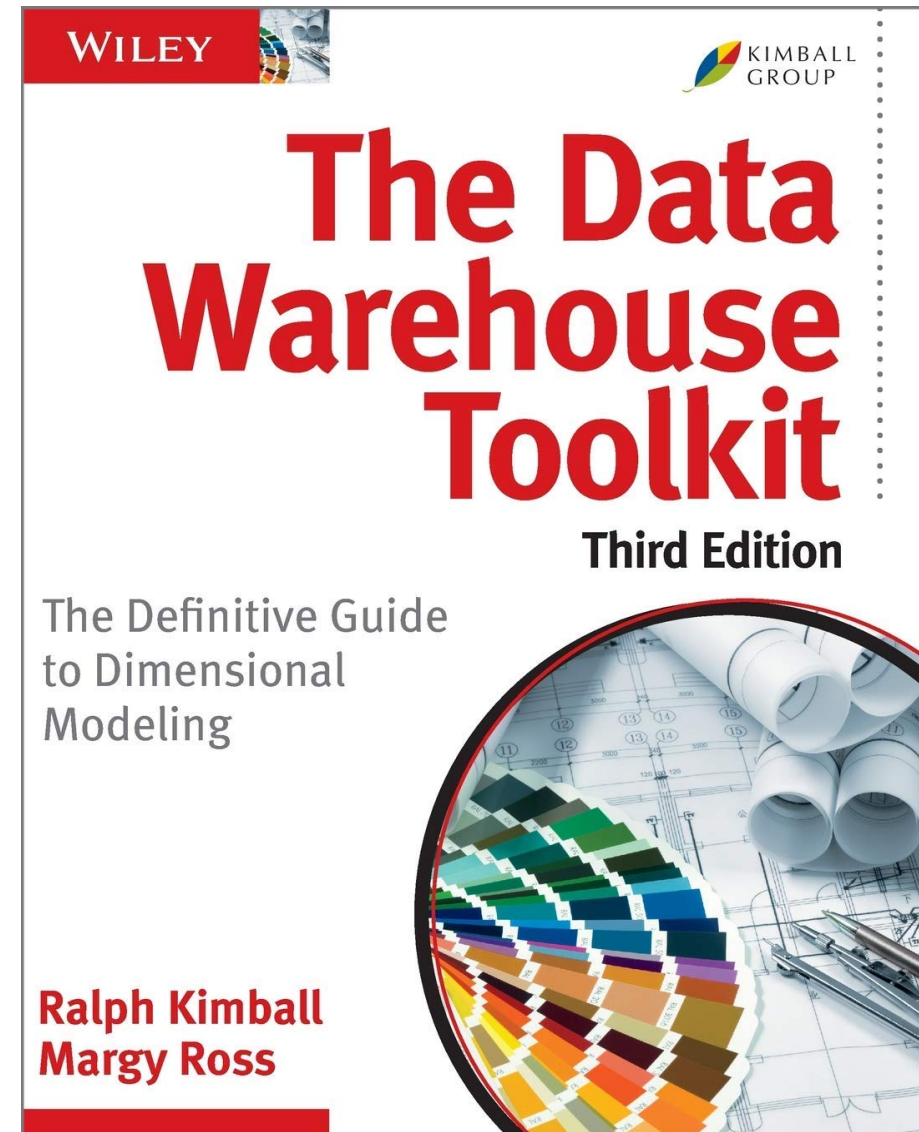


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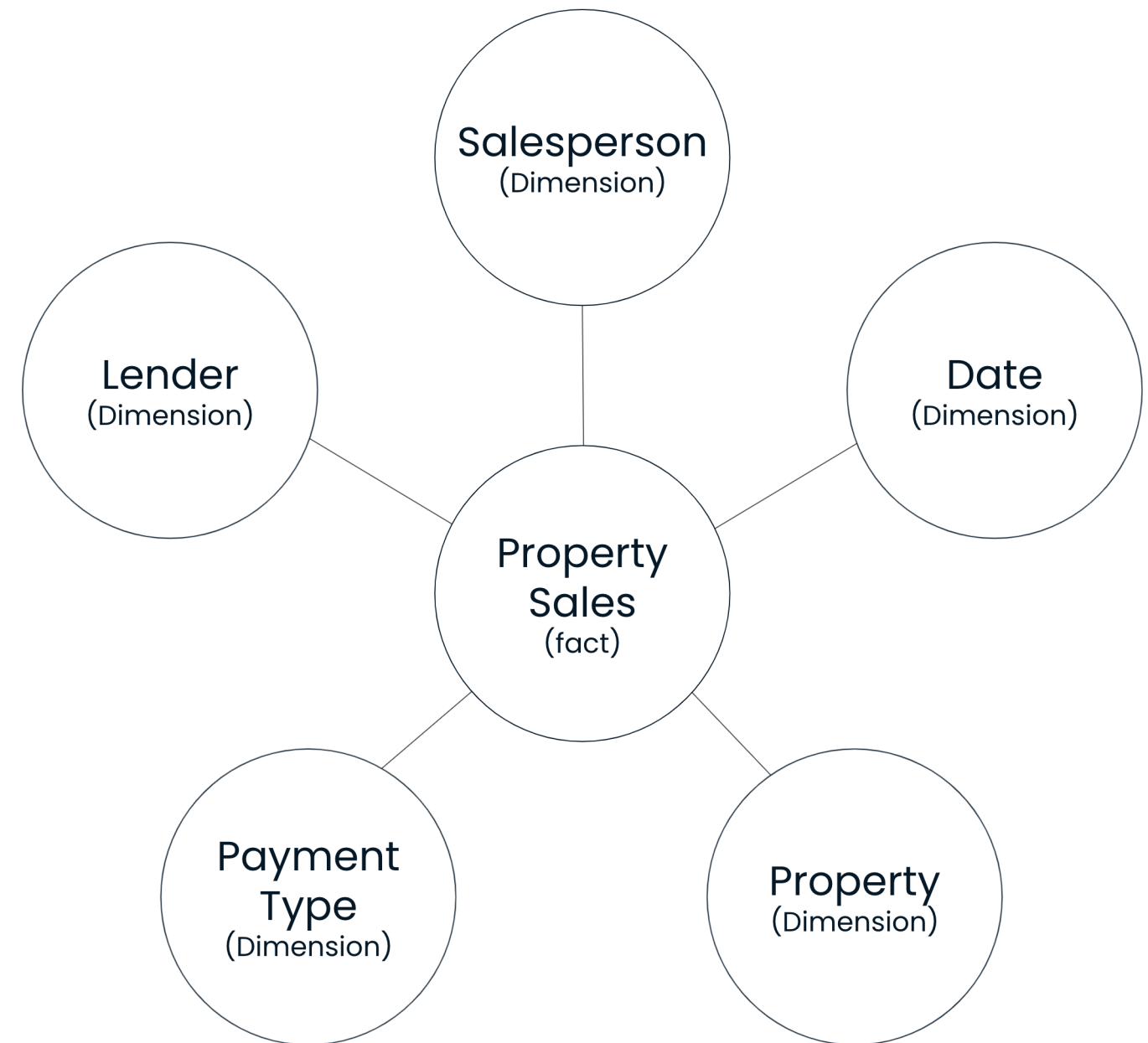
# The Kimball Model

The dimensional model



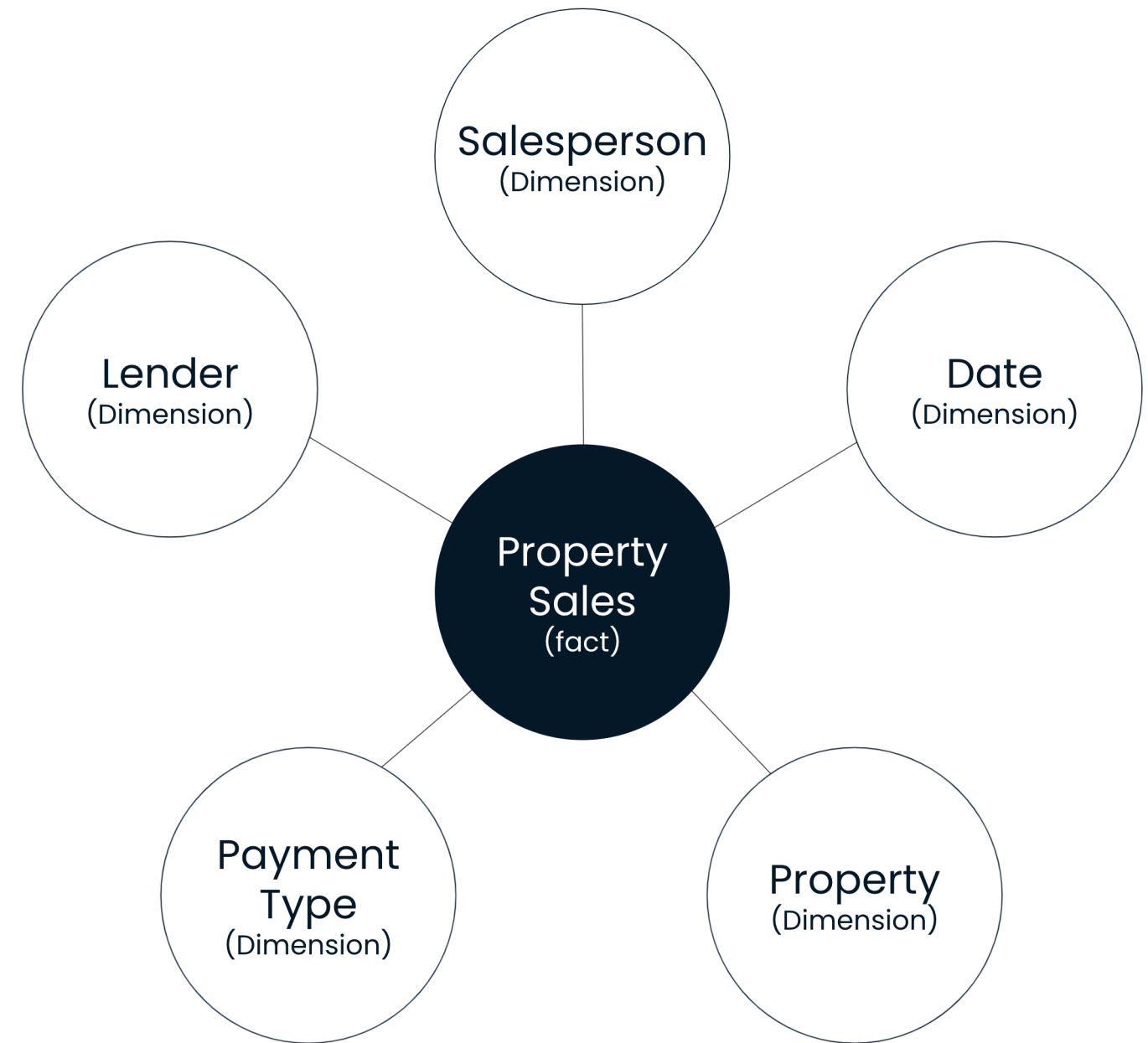
# The Kimball Model

- Key concepts
  - **Facts:** metrics from a business process
  - **Dimensions:** context surrounding a business process
  - Combine to form a **star schema**
- Star schemas are used in data warehouses
- Power BI is optimized for star schemas



# Fact tables

- Made up of
  - **Facts (measures)**
    - Measurements or metrics from your business process
  - **Keys**
    - Used to establish relationships between fact and dimension tables
- Fact tables are long and narrow
  - Lots of rows
  - Fewer columns



# Fact tables: an example

Property Sales table

| LenderID | StartDateID | PropertyID | PaymentTypeID | SalesPersonID | Rent | Duration |
|----------|-------------|------------|---------------|---------------|------|----------|
| CO76     | 20200624    | PG14       | P2            | SA9           | 750  | 24       |
| CO56     | 20200907    | PG4        | P4            | SA12          | 1250 | 12       |
| CO62     | 20201201    | PG16       | P3            | SA5           | 3000 | 36       |
| CO43     | 20200201    | PG6        | P3            | SA6           | 500  | 24       |
| CO76     | 20200530    | PG20       | P2            | SA6           | 5000 | 12       |
| CO76     | 20200115    | PG11       | P2            | SA2           | 2000 | 24       |
| CO32     | 20201201    | PG15       | P2            | SA9           | 450  | 36       |
| ...      | ...         | ...        | ...           | ...           | ...  | ...      |

# Fact tables: an example

Keys: establish relationships between tables

| LenderID | StartDateID | PropertyID | PaymentTypeID | SalesPersonID | Rent | Duration |
|----------|-------------|------------|---------------|---------------|------|----------|
| CO76     | 20200624    | PG14       | P2            | SA9           | 750  | 24       |
| CO56     | 20200907    | PG4        | P4            | SA12          | 1250 | 12       |
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| CO43     | 20200201    | PG6        | P3            | SA6           | 500  | 24       |
| CO76     | 20200430    | PG20       | P2            | SA9           | 5000 | 12       |
| CO76     | 20200115    | PG11       | P2            | SA2           | 2000 | 24       |
| CO32     | 20201201    | PG15       | P2            | SA9           | 450  | 36       |
| ...      | ...         | ...        | ...           | ...           | ...  | ...      |

# Fact tables: an example

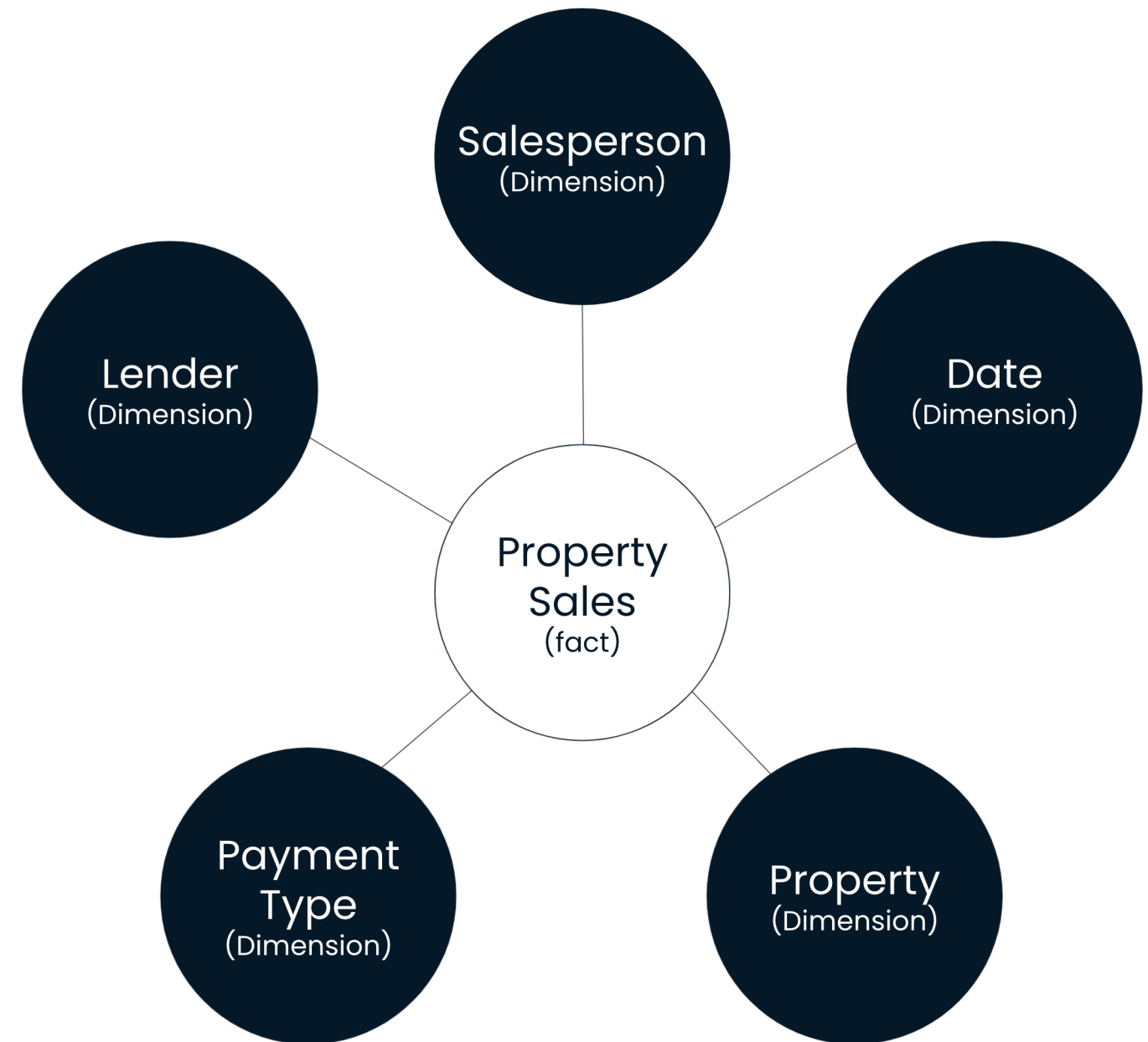
**Measures:** metrics from the business process

| LenderID | StartDateID | PropertyID | PaymentTypeID | SalesPersonID | Rent | Duration |
|----------|-------------|------------|---------------|---------------|------|----------|
| CO76     | 20200624    | PG14       | P2            | SA9           | 750  | 24       |
| CO56     | 20200907    | PG4        | P4            | SA12          | 1250 | 12       |
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| ...      | ...         | ...        | ...           | ...           | ...  | ...      |



# Dimension tables

- Provide context
  - Who, what, when, where, why?
- Shared business concepts
  - E.g., person, employee, customer, vendor
- Contain static or "slowly changing" data
  - E.g., name, date of birth, height
- Dimension tables are short and wide
  - Few rows
  - Lots of columns





# Dimension tables: an example

Salesperson table

| SalesPersonID | FirstName | LastName | DateOfBirth | Salary |
|---------------|-----------|----------|-------------|--------|
| SA9           | Mary      | Howe     | 1990-02-19  | 24000  |
| SA12          | David     | Ford     | 1978-03-24  | 18000  |
| SA5           | Ann       | Beech    | 1980-11-10  | 12000  |
| SA6           | Julie     | Lee      | 1985-06-13  | 30000  |
| SA9           | John      | White    | 1965-10-01  | 9000   |
| ...           | ...       |          | ...         |        |

# Dimension tables: an example

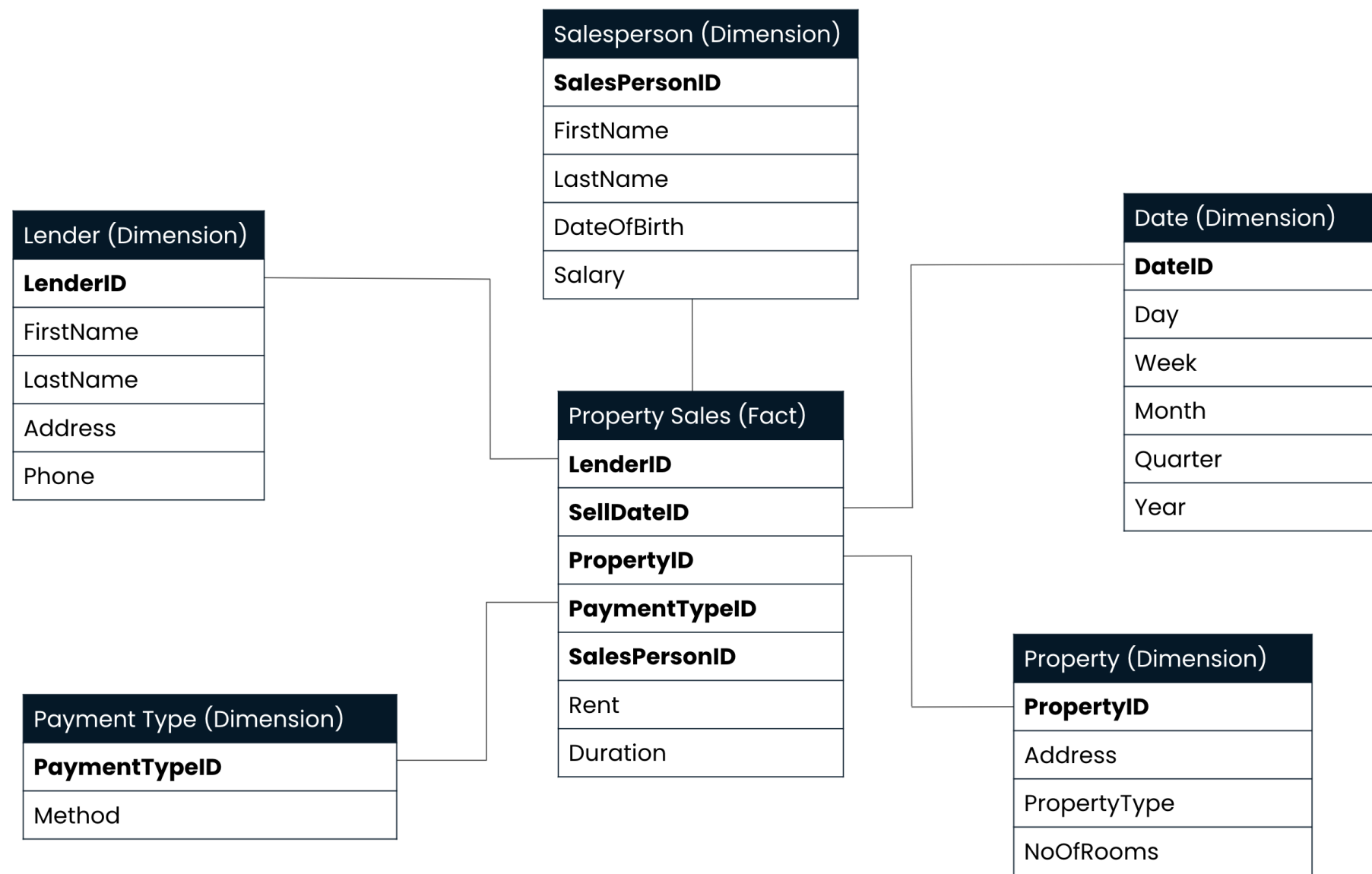
Key: establish relationship with fact table

| SalesPersonID | FirstName | LastName | DateOfBirth | Salary |
|---------------|-----------|----------|-------------|--------|
| SA9           | Mary      | Howe     | 1990-02-19  | 24000  |
| SA12          | David     | Ford     | 1978-03-24  | 18000  |
| SA5           | Ann       | Beech    | 1980-11-10  | 12000  |
| SA6           | Julie     | Lee      | 1985-06-13  | 30000  |
| SA9           | John      | White    | 1965-10-01  | 9000   |
| ...           | ...       |          | ...         |        |

# Dimension tables: an example

**Attributes:** various characteristics of the dimension

| SalesPersonID | FirstName | LastName | DateOfBirth | Salary |
|---------------|-----------|----------|-------------|--------|
| SA9           | Mary      | Howe     | 1990-02-19  | 24000  |
| SA12          | David     | Ford     | 1978-03-24  | 18000  |
| SA5           | Ann       | Beech    | 1980-11-10  | 12000  |
| SA6           | Julie     | Lee      | 1985-06-13  | 30000  |
| SA9           | John      | White    | 1965-10-01  | 9000   |
| ...           | ...       | ...      | ...         | ...    |



- Dimensions are used in multiple facts
- Dimensions do not link to other dimensions



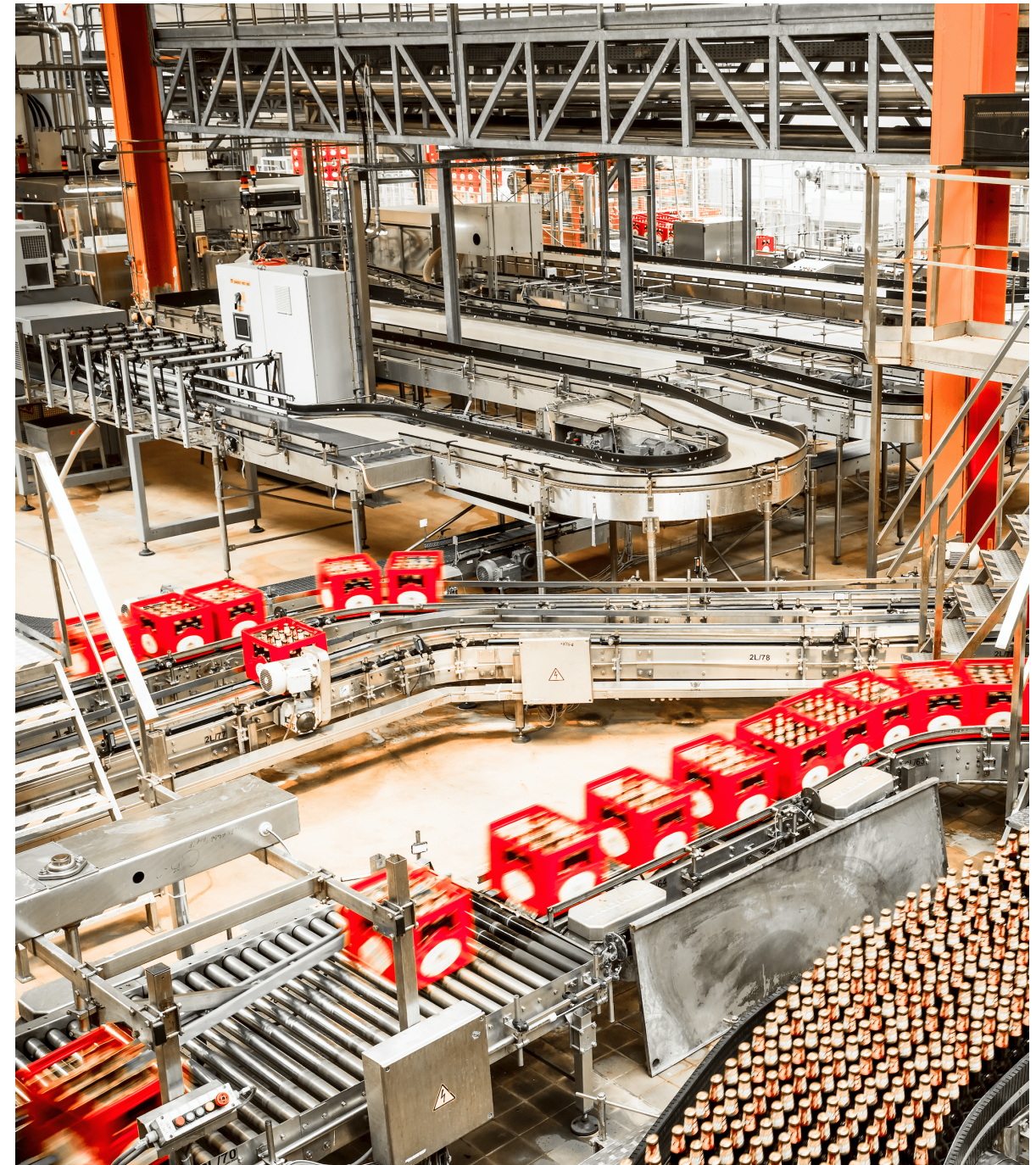
# The dataset

## Fact

- **Establishment Survey:** number of employees, number of firms, ...

## Dimensions

- **Industry:** NAICS code, industry group, subsector, sector
- **Time:** year, decade, century
- **Age:** establishment age
- **Geography:** country, state



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# Creating a star schema

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# Let's practice!

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